MIL-P-24191C(SH)
31 March 1981
SUPERSEDING
MIL-P-24191B(SHIPS)
23 August 1973
(See 6.5)

MILITARY SPECIFICATION

PLASTIC SHEET, CAST, ACRYLIC, SHIPBOARD APPLICATION

(ILLUMINATION AND SIGNAL LIGHTING)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cast acrylic sheet materials.

1.2 Classification.

* 1.2.1 Plastic sheet shall be furnished in the following types, grades, and thicknesses as specified (see 6.2, tables I and III).

Type RTPS-1 - Red transparent.
Type RTPS-2 - Red transparent, smooth surface finish, one side; patterned pyramidal surface, second side.
Type YTPS-1 - Yellow transparent.
Type GTPS-1 - Green transparent.
Type BTPS-1 - Blue transparent.
Type BTPS-2 - Blue transparent, smooth surface finish, one-side; patterned pyramidal surface, second side.
Type WTLS-1 -
Type WTLS-2 - } White translucent (Grades A through G (see table
Type WTLS-3 - } III)).
Type WTLS-4 -
Type RTLS-1 - Red translucent.
Type CLEAR-1 - Clear (colorless).
Type CLEAR-2 - Clear (colorless), smooth surface finish, one side; patterned pyramidal surface, second side.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 White translucent sheet shall be furnished in the type and grade specified in table I (see 6.2.1). The thickness in table I is a requirement of the type and grade specified.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 256-78E - Impact Resistance of Plastics and Electrical Insulating Materials, Tests for.
D 542-50R70 - Index of Refraction of Transparent Organic Plastics, Tests for.
D 570-77 - Water Absorption of Plastics, Test for.
D 635-77 - Rate of Burning and/or Extent and Time of Burning of Self-Supporting Plastics in a Horizontal Position, Test for.
D 648-72 - Deflection Temperature of Plastics Under Flexural Load, Test for.
D 788-78 - Methacrylate Molding and Extrusion Compounds.
D 790-71 - Flexural Properties of Plastics and Electrical Insulating Materials, Tests for.
D 792-66R75 - Specific Gravity and Density of Plastics by Displacement, Tests for.
D 1003-61R70 - Haze and Luminous Transmittance of Transparent Plastics, Tests for.
D 1925-70R77 - Yellowness Index of Plastics, Tests for.
D 2843-77 - Density of Smoke from the Burning of Decomposition of Plastics, Tests for.
E 167-63R70 - Goniophotometry of Object and Materials, Rec. Practice for.
E 308-66 - Spectrophotometry and Description of Color in CIE 1931 System, Rec. Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material compliance. Prior to submission of finished items (see 6.1) for acceptance, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the flat sheet materials, prior to forming into specific shapes of window assemblies as specified in the equipment specification for lighting fixtures for Naval Service, comply with all requirements of this specification. Satisfactory evidence shall consist of the submission of test reports of the inspection specified in 3.1.1 and 3.1.2 or notarized certification to this effect. If certification is submitted, this certification shall contain a statement to the effect that test records are available upon request at any time within a time frame of not less than one year.

* 3.1.1 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.5).

3.1.2 Quality conformance inspection. Quality conformance inspection during production shall consist of the inspection specified in 4.5.

3.2 Materials. The material shall be methacrylate base plastic, manufactured and processed so as to conform to all the requirements of this specification.

* 3.2.1 Surface finish for all types except for RTPS-2, BTPS-2 and CLEAR-2. Flat sheets shall have a smooth surface finish on both sides.

* 3.2.2 Surface finish for RTPS-2, BTPS-2 and CLEAR-2. Type RTPS-2, BTPS-2 and CLEAR-2 flat sheets shall have a smooth surface finish on one side and a patterned pyramidal surface on the other side which shall consist of a uniform array of 64 square pyramidal prisms per square inch, each with an included angle of 118 ± 2 degrees at the apex and approximately 0.021 inch high.

3.3 Dimensions.

3.3.1 Length and width. Length and width of sheets shall be as specified in the contract or order (see 6.1 and 6.2.1).

* 3.3.2 Thickness. Thickness of flat sheets that are used to fabricate window assemblies of specific shape as specified in the equipment specifications of lighting fixtures shall be as specified in the contract or order to suit contractor's fabrication procedures (see 6.1 and 6.2). Unless otherwise specified in the equipment specifications for lighting fixtures, thickness tolerance of flat sheets which are only cut to specified dimensions without further forming to specific shapes shall be as shown in table I. Thickness of type RTPS-2, BTPS-2 and CLEAR-2 shall be measured from the flat smooth surface to the vertex of the prisms.

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* TABLE I. Nominal thickness, thickness tolerance and formability diameter.

Nominal thickness (inch)[1]	Thickness tolerances (inch)		Formability diameter (inches) (heated)
	All types and grades except RTPS-2, BTPS-2 and CLEAR-2	RTPS-2, BTPS-2 and CLEAR-2[3] only	
[2]0.060	+0.015 -0.019	-	1.5
0.080	+0.014 -0.020	-	2.0
0.100	+0.013 -0.021	-	2.5
0.125	+0.015 -0.025	+0.045 -0.015	3.1
0.150	+0.019 -0.031	-	3.8
0.187	+0.017 -0.033	+0.043 -0.027	4.7
0.220	+0.020 -0.040	-	5.5
0.250	+0.020 -0.040	+0.045 -0.020	6.3

[1] 0.030, 0.100, 0.150, and 0.220 inch nominal thickness not available for type WTLS-1 through 4.
[2] 0.060 inch nominal thickness not available in types GTPS-1, BTPS-1, and RTPS-1.
[3] Not available in thicknesses 0.060, 0.080, 0.100, 0.150, and 0.220.

* TABLE II. Flammability characteristics.

Characteristic	Test specimen (inches)[1]	Applicable test method	Limit
Flammability rate	5 length x 1/2 width x 1/4 thickness	ASTM D 635	1.35 maximum inches/minute
Smoke density rating	1 length x 1 width x 1/4 thickness	ASTM D 2843	50 maximum

[1] A test specimen shall be made to the dimensions shown form the same batch of material offered for delivery.

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TABLE III. Luminous transmittance of white translucent sheet.

Type	Nominal thickness	Luminous transmittance - percent																			
		Grade A			Grade B			Grade C			Grade D			Grade E			Grade F			Grade G		
	Inch	Min.	Nominal	Max.	Min.	Nominal	Max.	Min.	Nominal	Max.	Min.	Nominal	Max.	Min.	Nominal	Max.	Min.	Nominal	Max.	Min.	Nominal	Max.
WTLS-1	0.060	42.0	47.0	52.0	27.0	31.0	35.0	75.0	84.0	93.0	72.0	81.0	90.0	63.0	71.0	79.0	82.0	87.0	92.0	63.0	70.0	77.0
WTLS-2	0.125	29.0	33.0	37.0	14.0	18.0	22.0	66.0	74.0	82.0	58.0	65.0	72.0	49.0	55.0	61.0	76.0	81.0	86.0	63.0	70.0	77.0
WTLS-3	0.187	20.0	24.0	28.0	10.0	13.0	16.0	58.0	65.0	72.0	48.0	54.0	60.0	39.0	44.0	49.0	72.0	77.0	82.0	63.0	70.0	77.0
WTLS-4	0.250	15.0	19.0	23.0	8.0	10.0	12.0	48.0	54.0	60.0	40.0	45.0	50.0	31.0	35.0	39.0	67.0	72.0	77.0	63.0	70.0	77.0

TABLE IV. Optical property values for clear (colorless) sheets.

Property	Value	ASTM test method
Index of refraction	1.49 min., 1.50 max.	ASTM D 542 Refractometric method
Haze	3.0 percent max.	ASTM D 1003
Light transmittance 0.060 - 0.187 in. thick 0.220 - 0.250 in. thick	 91 percent min. 87 percent min.	ASTM E 308
Spectral transmittance at any wave length in the 290 to 330 namometers (nm) band with 0.250 in. thick sheet.	5 percent max.1/	ASTM D 1003 Using procedure B

1/ When testing other thickness, the value found shall be adjusted to 0.250 inch thickness.

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3.4 Formability. When heated in accordance with the instructions furnished by the manufacturer, the sheets shall be suitable for bending into cylindrical forms of the diameter specified in table I without the appearance of crazing or other surface irregularities.

3.5 Optical uniformity. Plastic sheet shall be free from defects such as imbedded particles, bubbles, scratches, blemishes, and striations, such as will affect the optical properties as specified in 3.5.1 through 3.5.3.

* 3.5.1 Chips. Maximum allowable size shall be 1/8 inch. Chips approximately 1/8 inch in size shall not have a frequency greater than one chip per 4 square feet of sheet area. Chips less than 1/8 inch shall not form a concentrated pattern that may affect serviceability. Chips out-of-tolerance in size may be knifed off and considered acceptable if the remaining blemish could be removed by polishing.

* 3.5.2 Embedded particles. Maximum allowable dimension shall be 1/8 inch. Dirt less than 1/32 inch shall be disregarded. The maximum permissible occurrence of dirt with dimensions ranging from 1/32 to 1/8 inch shall be 1 per 4 square feet of sheet area.

* 3.5.3 Other defects. Minor defects which can be removed by polishing will be acceptable. Defects witnin one inch of the untrimmed edge of the sheet, which do not significantly reduce mechanical strength of the sheet, shall be permitted.

3.6 Specific gravity. The specific gravity of the material shall be 1.18 to 1.20.

3.7 Deflection temperature under load. The deflection temperature under load shall be not less than 185°F.

3.8 Water absorption. The water absorption shall not exceed the limits specified on figure 1 for the appropriate thickness.

* 3.9 Flammability. Flammability rate and smoke density rating shall not exceed the limits specified in table II.

3.10 Flexural strength. The flexural strength at break shall not be less than 14,000 pounds per square inch (lb/in^2).

3.11 Impact strength (Izod). The Izod impact strength shall be not less than 0.25 foot pounds per inch of notch.

3.12 Optical properties.

3.12.1 Color (chromaticity) and luminous transmittance. The chromaticity in CIE[1/] coordinates and luminous transmittance for the types specified herein shall be as specified in 3.12.1 to 3.12.3, inclusive.

1/ Commission Internationale de I'Eclairage (International Commission on Illumination).

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* 3.12.1.1 Types RTPS-1 and RTPS-2. Red transparent color shall fall within the area bounded by the spectrum locus and y = 0.300, x = 0.985-y as shown on figure 2. Luminous transmittance, using CIE illuminant A, shall be not less than 6 percent when specimens are tested as specified in 4.6.8. Test samples for RTPS-2 shall be prepared as specified in 4.6.1.3.

3.12.1.2 Type YTPS-1. Yellow transparent color shall fall within the areas bounded by the spectrum locus and

y = 0.382	x = 0.454 + 0.240y
y = 0.790-0.667x	x = 0.698-0.333y

as shown on figure 2. Luminous transmittance, using CIE illuminant A, shall be not less than 50 percent when specimens are tested as specified in 4.6.8.

* 3.12.1.3 Type GTPS-1. Green transparent color shall fall within the area bounded by the spectrum locus and

y = 0.390-0.171x	x = 0.360-0.080y
	x = 0.650y-0.030

as shown on figure 2. Luminous transmittance, using CIE illuminant A, shall be not less than 20 percent when specimens are tested as specified in 4.6.8

* 3.12.1.4 Types BTPS-1 and BTPS-2. Blue transparent color shall fall within the area bounded by the spectrum locus and

y = 0.230	x = 0.133+0.600y
	x = 0.400-y

as shown on figure 2. Luminous transmittance, using CIE illuminant A, shall be not less than 2 percent when specimens are tested as specified in 4.6.8. Test samples for BTPS-2 shall be prepared as specified in 4.6.1.3.

* 3.12.1.5 Type WTLS-1 through 4, grades A through G. The optical properties of type WTLS-1 through 4, grades A through G shall be as specified below:

(a) Luminous transmittance. Luminous transmittance shall be as specified in table III.
(b) Yellowness index. Yellowness index shall be 21 maximum.
(c) Diffusion factor. Diffusion factor shall be not less than 0.30.
(d) Filament hiding power. Filament hiding power shall be not less than 0.20.

* 3.12.1.6 Type RTLS-1. Red translucent color shall fall within the area bounded by the spectrum locus and y = 0.335, x = 0.985-y, as shown on figure 2. Luminous transmittance, using CIE illuminant A, shall be not less than 10 percent when tested as specified in 4.6.8.

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* 3.12.2 Type CLEAR-1. Optical properties for type CLEAR-1 plastic sheets shall be as specified in table IV. The light transmitted by clear plastic sheets shall not be noticeably different in chromaticity from that of the illuminant.

* 3.12.3 Type CLEAR-2. Optical properties for type CLEAR-2 plastic sheets shall be as specified in table IV except that the test sample shall be prepared as specified in 4.6.1.3.

3.13 Instruction sheet. An instruction sheet shall be furnished with each shipment. Specific information or reference shall be made to the limitations of the material and the necessary precautions to be observed in handling, storing, cutting, drilling, machining, forming, bending, cementing, abrading, polishing, and cleaning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

* 4.3 First article inspection. The sample for first article inspection shall be subjected to the tests of table V.

TABLE V. First article inspection tests.

Tests	Requirement paragraph	Test paragraph
Dimensional examination	3.2, 3.3, 3.5	4.5.1
Formability	3.4	4.6.9
Optical uniformity	3.5	4.5.1
Specific gravity	3.6	4.6.2
Deflection temperature under load	3.7	4.6.3
Water absorption	3.8	4.6.4
Flammability	3.9	4.6.5
Flexural strength	3.10	4.6.6
Impact strength	3.11	4.6.7
Optical properties	3.12	4.6.8

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4.3.1 First article inspection report. The contractor shall furnish a first article inspection report in accordance with the data ordering document included in the contract (see 6.2.2).

4.4 Sampling for quality conformance inspection.

4.4.1 Lot. All material of the same type, grade, and thickness manufactured under essentially the same conditions from the same batch of material and offered for delivery at the same time shall be considered a lot for purposes of inspection.

4.4.2 Sampling for visual and dimensional examination. A random sample of sheets shall be selected from each lot offered for examination in accordance with MIL-STD-105 at inspection level II. The acceptable quality level shall be equal to 1.5 percent defective.

4.4.3 Sampling for tests. A random sample of sheets shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-2, except that only one sheet shall be selected from lots of 40 or under. The random sample shall be subjected to the tests specified in 4.6. The acceptable quality level shall be equal to 1.5 percent defective.

4.5 Quality conformance inspection.

4.5.1 Visual and dimensional examination. Each sample sheet selected in accordance with 4.4.2 shall be subjected to visual and dimensional examination to determine compliance with 3.2, 3.3 and 3.5. If any sheet in the sample contains one or more defects, as defined in 3.5.1 and 3.5.2, it shall be rejected, and if the number of rejected sheets in the sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample. The thickness shall be determined by the use of a suitable instrument, accurate to 0.001 inch. Five measurements shall be made at random points on the sheet.

* 4.5.2 Quality conformance tests. Quality conformance tests shall be conducted on the sample sheets selected in accordance with 4.4.3 and shall consist of the tests specified in table VI. Results shall be averaged for each sample sheet to determine conformance with this specification. If the average for any sheet fails to conform with any requirement, this shall be cause for rejection of the lot. Color (chromaticity) and luminous transmittance do not apply to types RTPS-2, BTPS-2, and CLEAR-2.

TABLE VI. Quality conformance tests.

Test	Requirement paragraph	Test paragraph	Number of determinations
Specific gravity	3.6	4.6.2	1
Deflection temperature under load	3.7	4.6.3	3
Optical properties	3.12	4.6.8	1

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4.6 Test procedures.

4.6.1 Test samples.

4.6.1.1 Condition of test samples. Samples shall be tested in the "as received" condition; that is, no special conditioning shall be required unless otherwise specified in the test procedure.

4.6.1.2 Test thickness. Tests shall be made on sample sheets of nominal thickness shown in table I. If sample sheets are tested other than at the nominal thickness, the test results shall be adjusted to the nominal thickness.

* 4.6.1.3 Test samples for types RTPS-2, BTPS-2, and CLEAR-2. Test samples for RTPS-2, BTPS-2, and CLEAR-2 shall be prepared by either of the following methods:

(a) The patterned surface shall be ground down to the nearest nominal thickness shown in table I and polished to a smooth surface.
(b) The test specimen shall be cast with a smooth surface finish on both sides to the nearest nominal thickness shown in table I from the same batch material as that used for the production sheets under test.

4.6.2 Specific gravity. The specific gravity of the material shall be determined in accordance with ASTM D 792.

4.6.3 Deflection temperature under load. Conditioned test specimens shall be tested. The length of the specimen shall be 5 inches, the depth shall be 0.5 ± 0.005 inch, and the width shall be the thickness of the original sheet. Thin sheet materials, less than 0.125 inch (namely 0.060, 0.080 and 0.100 inch) shall be arranged to form a composite specimen of convenient thickness. The appratus shall be similar to ASTM D 648. The conditioned specimen shall be supported as a simple beam on steel supports 4 inches apart with the load applied vertically on the top of the specimen midway between the supports. The 1/2-inch dimension or depth of the beam shall be vertical. The load applied shall be governed by the width of the specimen and shall in each case induce in the specimen a maximum outer fiber stress of 264 lb/in^2. The loads applied for typical specimens are as follows:

Specimen	Load applied midway between the supports
1/4 by 1/2 by 5 inches	2.75 pounds
1/8 by 1/2 by 5 inches	1.38 pounds

The test specimens shall be in direct contact with a suitable liquid heat transfer medium, liquid petroleum USP, which is agitated during the test. The temperature of liquid shall be raised from normal room temperature at a rate of 3.6°F ± 0.9°F per minute. The heat distortion temperature shall be considered the temperature at which the specimen shall deflect 0.010 inch at the center between the supports.

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4.6.3.1 Sheets made by the extrusion process of the continuous cast. Extrusion process shall be tested for deflection temperature under load in accordance with the conditioning of test specimens specified for grade 8 material of ASTM D 788.

For 1/8 inch specimens: 185°F
For 1/4 inch specimens: 185°F

The two hour period shall be applicable to the sheets of 0.060, 0.080, 0.100 and 0.125 inch thickness. The four hour period shall be applicable to the sheets of 0.150, 0.187, 0.220 and 0.250 inch thickness.

4.6.4 Water absorption. The percentage of water absorbed in 24 hours shall be determined in accordance with the water absorption test (for weight and dimensional changes) of ASTM D 570.

* 4.6.5 Flammability. Tests shall be conducted in accordance with methods shown in table II.

4.6.6 Flexural strength (tested flatwise). The flexural strength shall be determined in accordance with ASTM D 790.

4.6.7 Impact strength. Test for impact strength shall be performed in accordance with ASTM D 256, method A. Specimens shall be tested edgewise.

4.6.8 Tests for optical properties. Tests for optical properties shall be as specified in 4.6.8.1 through 4.6.8.3.

* 4.6.8.1 Types RTPS-1, RTPS-2, YTPS-1, GTPS-1, BTPS-1, BTPS-2, and RTLS-1. Tests for color and luminous transmittance for types RTPS-1, RTPS-2, YTPS-1, GTPS-1, BTPS-1, BTPS-2, and RTLS-1 shall be conducted in accordance with ASTM E 308. The tests shall be conducted using illuminant A.

4.6.8.2 Type WTLS-1 through 4, Grades A through G. Tests shall be as specified in 4.6.8.2.1 through 4.6.8.2.3 using CIE illuminant C.

4.6.8.2.1 Luminous transmittance. Luminous transmittance tests shall be conducted in accordance with ASTM E 308 using the Hardy-type spectrophotometer. Other apparatus is satisfactory if equivalent results are obtained.

4.6.8.2.2 Yellowness index. Yellowness index tests shall be conducted in accordance with ASTM D 1925.

* 4.6.8.2.3 Diffusion factor and filament hiding power. Diffusion factor (DF) and filament hiding power (FHP) of a material, when illuminated by parallel light incident normally to the specimen, are defined by:

$$DF = \frac{B_{20} + B_{70}}{2B_5}$$

$$FHP = \frac{B_2}{B_0}$$

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Where:

B_0 = Brightness at the normal (straight through) view angle.
B_2 = Brightness at a view angle 2 degrees from normal.
B_5 = Brightness at a view angle 5 degrees from normal.
B_{20} = Brightness at a view angle 20 degrees from normal.
B_{70} = Brightness at a view angle 70 degrees from normal.

This specimen shall be viewed from the side opposite to that on which the light is incident, and measurements shall be made on a goniophotometer using appropriate instrumentation for recording brightness in accordance with ASTM E 167. Diffusion factor and filament hiding power shall be calculated using the above formulas.

* 4.6.8.3 Type CLEAR-1 and CLEAR-2. One specimen each shall be tested as specified in table IV.

4.6.9 Formabililty. The formability of the material shall be determined by heating the material in accordance with the supplier's instructions and by bending it into cylindrical forms of the diameter specified in table I.

5. PACKAGING

5.1 Not applicable.

6. NOTES

6.1 Intended use. The flat plastic sheet covered by this specification is intended for use in fabrication of window assemblies for lighting fixtures for Naval service.

6.1.1 Reference to this specification on drawings or in product specifications may be made in order to specify the flat material to be used in the fabrication of a part. It should be clearly understood that the requirements apply to the flat material prior to forming or fabrication and not necessarily to the material in its final form.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, grade and thickness required (see 1.2, tables I and III).
(c) When a first article sample is required (see 3.1.1).
(d) Length, width and thickness required (see 3.3).

6.2.2 Data requirements. When this specification is used in a contract which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the

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data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirements	Applicable DID	Option
4.3.1	First Article Inspection Report	DI-T-4902	----

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

* 6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Fabricated parts or equipment. Requirements applicable to fabricated parts or equipment incorporating this material should be specified in separate specifications or as part of contract or orders.

6.4 Annealing. All parts fabricated from flat sheets should be heat annealed in accordance with specific instructions supplied by the manufacturer of the flat sheet.

* 6.5 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - SH
(Project 9330-N858)

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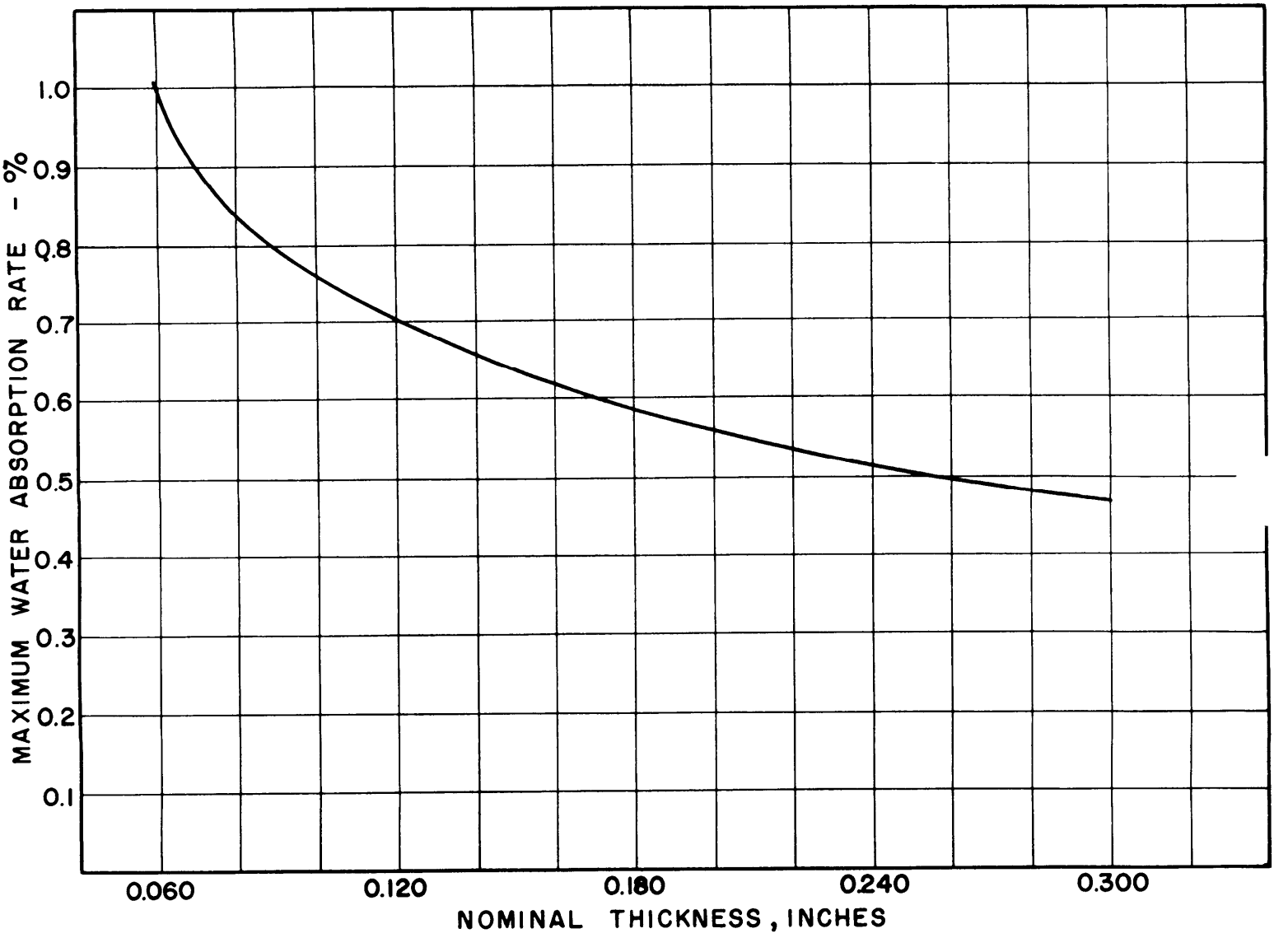


FIGURE 1. Water absorption limits.

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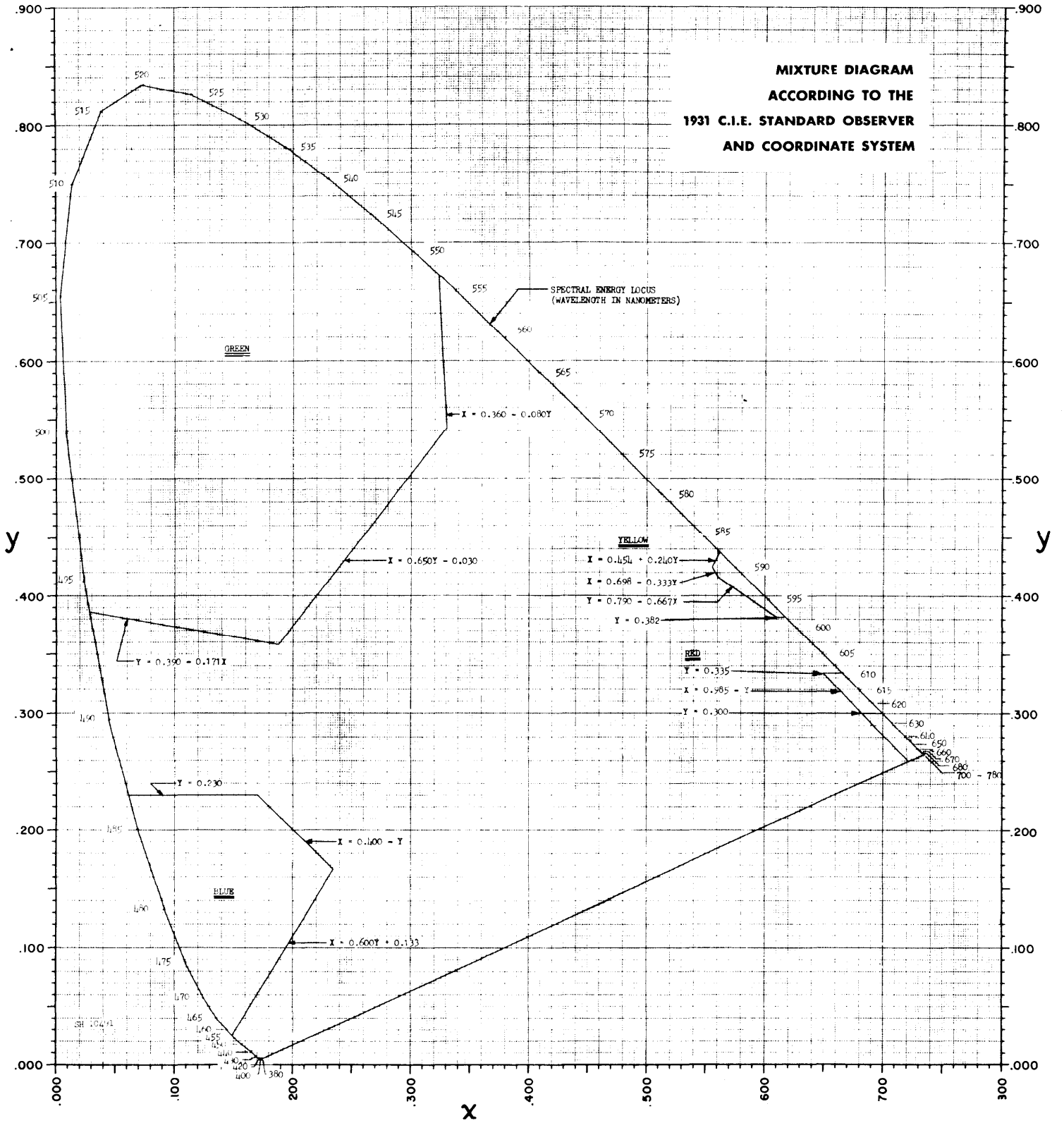


FIGURE 2. Chromaticity diagram illuminate A.

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